Email Spam Detection Using Machine Learning

Yash Talaviya (92410103105) , Tirth Bhut (92410103093), Pratham Gajra (92410103106), Prof. Yogeshwar Prajapati

Abstract

Email communication has become an essential part of modern life, supporting both personal and professional interactions. However, with the rapid growth in email usage, the number of spam and phishing attacks has also increased significantly. These malicious emails often contain misleading information, fake offers, or harmful links that can lead to data theft, financial loss, and serious security risks. As a result, there is a strong need for an efficient and automated system to identify and filter such emails. This project presents a machine learning-based system for detecting spam emails using advanced text processing and classification techniques. The system begins with the preprocessing of email content, including cleaning, normalisation, and removal of irrelevant data. It then applies TF-IDF (Term Frequency–Inverse Document Frequency) for feature extraction, which converts textual data into meaningful numerical representations. A Linear Support Vector Classifier (LinearSVC) is used as the primary model due to its effectiveness and fast performance in text classification tasks. Additionally, a calibration technique is applied to generate probability-based outputs, allowing the system to provide confidence scores for each prediction. The model is trained and tested on a large dataset containing both spam and legitimate emails. Its performance is evaluated using standard metrics such as accuracy, precision, recall, and F1-score. Experimental results demonstrate that the proposed system achieves high accuracy while maintaining low false positive and false negative rates. Furthermore, the system is capable of real-time classification, making it practical for deployment in real-world environments. Overall, the proposed approach offers a reliable, efficient, and scalable solution for email spam detection. It can be integrated into email platforms, cybersecurity systems, and web applications to enhance protection against spam and phishing attacks while improving user experience.